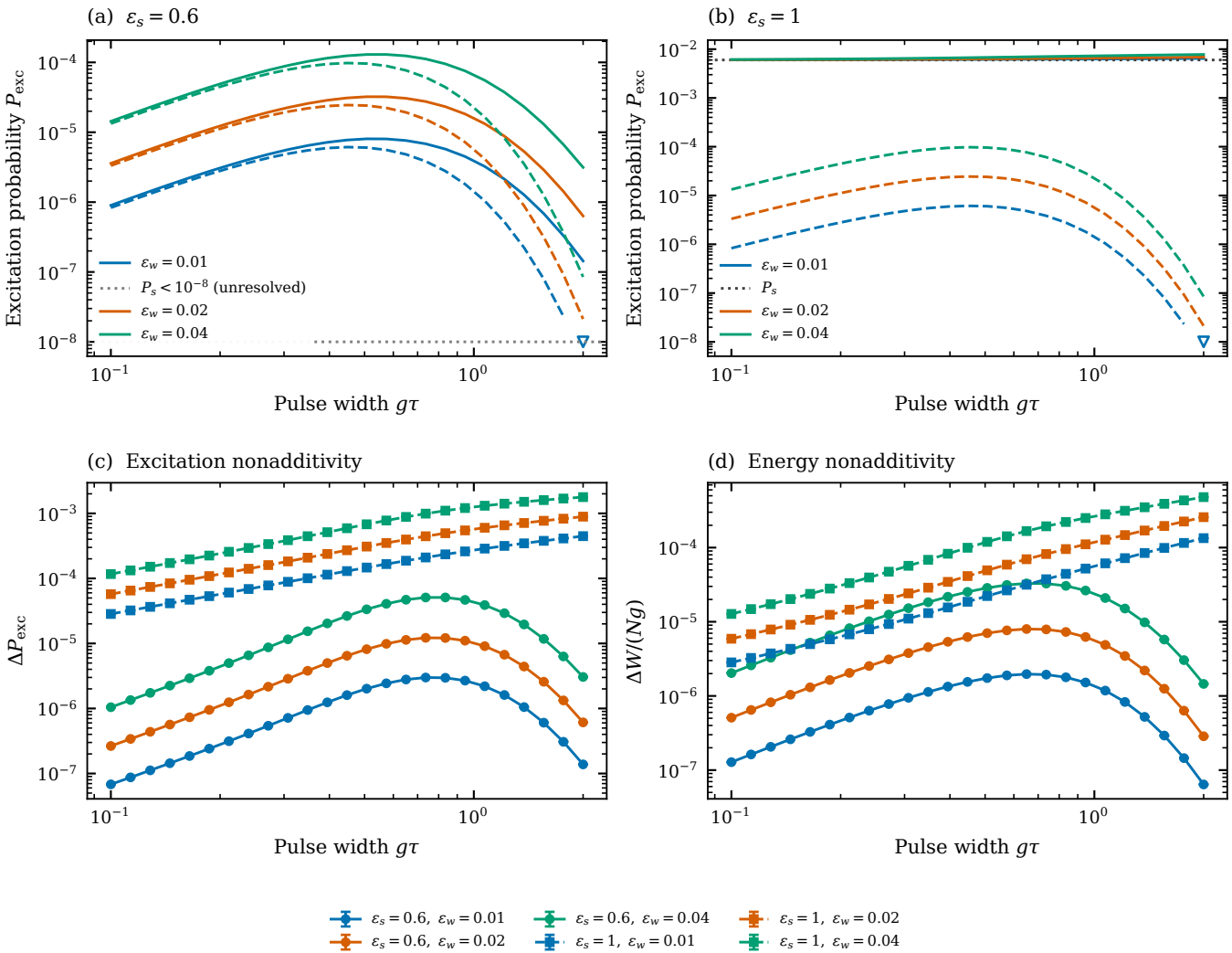

(a) $\varepsilon_s = 0.6$
Excitation probability $P_{\rm exc}$
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
$\varepsilon_w = 0.01$
$P_s < 10^{-8}$ (unresolved)
$\varepsilon_w = 0.02$
$\varepsilon_w = 0.04$
10^{-1}
10^{0}
Pulse width $g\tau$
(b) $\varepsilon_s = 1$
Excitation probability $P_{\rm exc}$
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
$\varepsilon_w = 0.01$
P_s
$\varepsilon_w = 0.02$
$\varepsilon_w = 0.04$
10^{-1}
10^{0}
Pulse width $g\tau$
(c) Excitation nonadditivity
$\Delta P_{\rm exc}$
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-1}
10^{0}
Pulse width $g\tau$
(d) Energy nonadditivity
$\Delta W/(Ng)$
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-1}
10^{0}
Pulse width $g\tau$
$\varepsilon_s = 0.6,\ \varepsilon_w = 0.01$
$\varepsilon_s = 0.6,\ \varepsilon_w = 0.02$
$\varepsilon_s = 0.6,\ \varepsilon_w = 0.04$
$\varepsilon_s = 1,\ \varepsilon_w = 0.01$
$\varepsilon_s = 1,\ \varepsilon_w = 0.02$
$\varepsilon_s = 1,\ \varepsilon_w = 0.04$